

Patched versions of Theorems 3.2.1 and 3.2.3, and the implication $3.2.1' \Rightarrow 3.2.3'$

Patched Lifting Theorem

Theorem 3.2.1' (Generalized Lifting Theorem). Let $f(x, x_r) \in \mathbb{R}[x, x_r]$ be squarefree, of positive degree in x_r . Let S be a connected submanifold of $\mathbb{R}^{r-1}$ in which f is degree-invariant and of non-negative degree. Suppose that some nonzero

$$P \in \langle f, \partial f / \partial x_r \rangle \cap \mathbb{R}[x]$$

is order-invariant in S . Then f is analytically delineable on S and is order-invariant in each f -section over S .

This generalizes the original 3.2.1 because $D = \text{disc}(f)$ is one particular nonzero element of $\langle f, f' \rangle \cap \mathbb{R}[x]$, so the original statement is a special case. The proof requires modifying McCallum's original argument (using Weierstrass preparation to handle the degree-drop case), but the structure is the same.

Generalized projection $P'(A)$

For a squarefree basis A , define

$$P'(A) = \text{coeff}(A) \cup \text{elim_self}(A) \cup \text{elim_pair}(A),$$

where

- $\text{elim_self}(A) = \{d_F : F \in A, \deg(F) \geq 2\}$, with each d_F a chosen nonzero element of $\langle F, F' \rangle \cap \mathbb{R}[x]$;
- $\text{elim_pair}(A) = \{r_{F,G} : F, G \in A, F \neq G, \deg(F), \deg(G) \geq 1\}$, with each $r_{F,G}$ a chosen nonzero element of $\langle F, G \rangle \cap \mathbb{R}[x]$.

This generalizes $P(A)$: taking $d_F = \text{disc}(F)$ and $r_{F,G} = \text{Res}(F, G)$ recovers the original.

Patched Projection Theorem

Theorem 3.2.3' (Generalized Projection Theorem). Let A be a finite squarefree basis of r -variate integral polynomials, $r \geq 2$, and let S be a connected submanifold of $\mathbb{R}^{r-1}$. Suppose each element of A is not identically zero on S , and each element of $P'(A)$ is order-invariant in S . Then each element of A is degree-invariant and analytically delineable on S , the sections of A over S are pairwise disjoint, and each element of A is order-invariant in every section of A over S .

Proof that 3.2.1' implies 3.2.3'

Let $F_1, \dots, F_n$ be the nonconstant elements of A , and set $f = \prod_{i=1}^n F_i$. Then f is squarefree (the F_i are squarefree and pairwise coprime). By hypothesis we have nonzero order-invariant elements

$$d_i \in \langle F_i, F'_i \rangle \cap \mathbb{R}[x], \quad r_{ij} \in \langle F_i, F_j \rangle \cap \mathbb{R}[x].$$

Set

$$R = \prod_i d_i \cdot \prod_{i < j} r_{ij}.$$

By Lemma 3.2.2, R is order-invariant on S (a product of order-invariant functions). And $R \neq 0$ since each factor is nonzero.

Claim

Some positive power R^N belongs to $\langle f, f' \rangle \cap \mathbb{R}[x]$.

Proof. Over $\bar{K}$ (algebraic closure), the repeated-root locus of f decomposes as

$$V(f, f') = \bigcup_i V(F_i, F'_i) \cup \bigcup_{i < j} V(F_i, F_j).$$

This is because $f = \prod F_i$ with F_i pairwise coprime and squarefree: a point (a, t) is a repeated root of f iff either some F_i has a repeated root there, or two distinct factors F_i, F_j vanish simultaneously. (For the reverse direction: if $F_i(a, t) = 0$ and no other $F_j(a, t) = 0$, then $f'(a, t) = G_i(a, t) \cdot F'_i(a, t)$, where $G_i = \prod_{j \neq i} F_j$, and $G_i(a, t) \neq 0$, so $f'(a, t) = 0$ forces $F'_i(a, t) = 0$.)

Projecting to $\bar{K}^{r-1}$,

$$\pi(V(f, f')) = \bigcup_i \pi(V(F_i, F'_i)) \cup \bigcup_{i < j} \pi(V(F_i, F_j)).$$

Since $d_i \in \langle F_i, F'_i \rangle \cap K[x]$, the polynomial d_i vanishes on $\pi(V(F_i, F'_i))$. Similarly r_{ij} vanishes on $\pi(V(F_i, F_j))$. Therefore

$$R = \prod_i d_i \cdot \prod_{i < j} r_{ij}$$

vanishes on the entire $\pi(V(f, f'))$.

Since $V(R)$ is Zariski-closed and contains $\pi(V(f, f'))$, it also contains the closure:

$$V(R) \supseteq \overline{\pi(V(f, f'))} = V(\langle f, f' \rangle \cap \bar{K}[x]).$$

By the Hilbert Nullstellensatz,

$$R \in \sqrt{\langle f, f' \rangle \cap \bar{K}[x]},$$

so $R^N \in \langle f, f' \rangle \cap \bar{K}[x]$ for some $N \geq 1$. Since f, f' , and R all have coefficients in $\mathbb{Z}$, this membership descends to $\mathbb{R}[x]$ (by flatness of the field extension, or equivalently because a Gröbner basis computation over $\mathbb{Q}$ gives the same elimination ideal). $\square$

Conclusion

R^N is a nonzero element of $\langle f, f' \rangle \cap \mathbb{R}[x]$. It is order-invariant on S : each d_i and r_{ij} is order-invariant, so R is order-invariant by Lemma 3.2.2, and

$$\text{ord}_a(R^N) = N \cdot \text{ord}_a(R)$$

is constant on S .

Furthermore, f is not identically zero on S and is degree-invariant on S (the coefficients of each F_i are order-invariant by hypothesis on $\text{coeff}(A)$).

Applying Theorem 3.2.1' to f with $P = R^N$: f is analytically delineable on S and order-invariant in each f -section over S . By Lemma 3.2.2, each F_i is order-invariant in every f -section. The sections of A over S are pairwise disjoint (since f is squarefree and delineable, distinct coprime factors cannot share a root on any section). $\square$

What this buys you

The generalized projection $P'(A)$ lets you replace $\text{disc}(F)$ by any nonzero element of $\langle F, F' \rangle \cap K[x]$, and $\text{Res}(F, G)$ by any nonzero element of $\langle F, G \rangle \cap K[x]$. If you can find simpler generators of these elimination ideals (e.g. via extended GCD in $K(x)[t]$ and clearing denominators), you get a smaller projection set. The Nullstellensatz bridges the gap: the product of these simpler elements, raised to some power, lands in $\langle f, f' \rangle \cap K[x]$, which is all the generalized lifting theorem needs.